

Feature Selection Visual Map

Climate metrics × phenological windows — which combinations were selected and at what resolution

 Coverage Heatmap

WINDOW POSITIONS

B1/B2/B3 — before phenostage (1×, 2×, 3× window back)
C — centred on phenostage ±n/2 days
A1/A2 — after phenostage (1×, 2× forward)

YEARS

Y1 = current (harvest) year
Y0 = previous year

CALCULATION TYPE

A — Average over window
S — Sum over window
C — Count (days meeting threshold)

CELL VALUE

Number = window length in days
Darker cell = longer window
Empty = not selected

Y1 · CURRENT YEAR — FEATURE HEATMAP

		 BUD BREAK					 FLOWERING					 START MATURITY					 HARVEST						
Metric		B2	B1	C	A1	A2	B3	B2	B1	C	A1	A2	A3	B2	B1	C	A1	A2	B2	B1	C	A1	A2
Mean Temp (tm)	A								10	15	10					20	20			20	10		
Days Tmed < 15°C	C								15	15	15												
Frost days (ti<0°C)	C			20	30						10												
Heat days (ta>35°C)	C										10	20			20		20						

SYMBOL	FULL NAME	CALC	UNIT	THRESHOLD	YEARS	DESCRIPTION
tm	Mean daily temperature	A Average	°C	—	Y1Y0	Daily mean temperature averaged over the window. A core thermal proxy for vine phenology: temperatures drive bud burst timing, berry set, sugar accumulation, and ripening speed.
tm<15°C	Cold days (Tmed < 15°C)	C Count	days	Tmed < 15°C	Y1	Count of days where mean temperature drops below 15°C. During flowering, cold conditions below this threshold impair pollen viability and fertilisation, directly reducing berry set and yield potential.
ti<0°C	Frost days (Tmin < 0°C)	C Count	days	Tmin < 0°C	Y1	Count of days with minimum temperature below 0°C. Frost events around bud break can irreversibly damage newly emerged shoots, causing partial or total yield loss for that growing season.
ta>35°C	Heat stress days (Tmax > 35°C)	C Count	days	Tmax > 35°C	Y1	Count of days with maximum temperature exceeding 35°C. Extreme heat during flowering causes flower abortion and poor set; during maturation it accelerates sugar rise while impairing acid retention and aroma.
rs	Rainfall sum	S Sum	mm	—	Y1	Total accumulated rainfall over the window period. Captures the overall water input from precipitation — critical around flowering (dilution risk) and harvest (disease, berry splitting).
rc>1mm	Wet days (rain > 1mm)	C Count	days	Rain > 1mm	Y1	Count of days with rainfall exceeding 1mm. Frequent wet days increase humidity and disease pressure (downy mildew, botrytis), and during flowering they physically hinder pollination.

SYMBOL	FULL NAME	CALC	UNIT	THRESHOLD	YEARS		DESCRIPTION
IAF	Leaf Area Index (IAF)	A Average	index	—	Y1	Y0	Índice de Área Foliar — solar radiation interception index averaged over the window. Reflects canopy size and photosynthetic capacity. High LAI before flowering indicates vigorous vegetative growth that can compete with reproductive development.
sw_a	Available soil water	A Average	mm	—	Y1	Y0	Mean soil water content available to the vine (between wilting point and field capacity). Sustained low values signal water stress; high values around harvest can reduce fruit concentration and quality.
<1.5×Wp	Severe water stress days	C Count	days	$Sm < 1.5 \times Wp$	Y1	Y0	Count of days where soil water falls below 1.5× the wilting point — a threshold for severe vine water stress. Douro: Wp = 402 mm; Vinho Verde: Wp = 338 mm. Prolonged stress reduces yield but can concentrate phenolics.
>0.9×Fc	Soil saturation days	C Count	days	$Sm > 0.9 \times Fc$	Y1	Y0	Count of days where soil water exceeds 90% of field capacity — indicating near-saturation conditions. Douro: Fc = 142.8 mm; Vinho Verde: Fc = 77 mm. Excess water limits root oxygen and increases disease risk.

PHENOLOGICAL STAGE		CALCULATION · WINDOW SIZE	
BB	Bud Break	10 Average – 10d	10 Sum – 10d
FL	Flowering	20 Average – 20d	20 Sum – 20d
SM	Start of Maturity	30 Average – 30d	10 Count – 10d
H	Harvest		20 Count – 20d